



INTRODUCTION

Arrowmark Vending is a small contract food services provider that has been awarded a contract to sell pizza at all Purdue University home football games during the 2026 season. This contract creates an operational decision for the company's operations manager, Tom Kealey, who must determine how many pizzas to produce for each game before demand is known. Because production must be completed in advance, overproducing results in waste and lost profit, while underproducing leads to missed sales and customer dissatisfaction. With demand at football games being highly variable, Tom must assess the expected profitability of each possible production level prior to each game. Tom has brought forth a list of cost and revenue assumptions as well as a range of production levels that he feels comfortable executing. He has tasked our team with analyzing these assumptions to help him optimize production levels.

1. PROBLEM STATEMENT AND ASSUMPTIONS

The purpose of this report is to determine the optimal number of pizzas of each type that Arrowmark Vending should produce before each game to maximize expected profit under uncertain demand. Production decisions are evaluated using the probability of each demand level occurring and the corresponding profit outcomes for each production quantity. By applying known demand distributions together with relevant cost and revenue information, this analysis provides Tom Kealey with data-driven guidance for selecting profit-maximizing pizza production levels for game-day operations.

Overview Of Our Assumptions:

This analysis calculates the Expected Monetary Value (EMV) for each feasible production and demand scenario to support profit-maximizing production decisions for the operations manager, Tom Kealey. The model evaluates the expected profitability of alternative production levels by incorporating both demand uncertainty and cost structure.

Several assumptions were used in constructing the models. Production quantities were assumed to range from 200 to 900 cheese pizzas and 300 to 800 pepperoni pizzas per game. Demand for each pizza type was assumed to be independent, with probabilities assigned to represent the likelihood of each demand level occurring. From a cost perspective, a fixed cost of \$1,000 per game was allocated evenly across cheese and pepperoni production (\$500 each). The variable cost of producing a cheese pizza was assumed to be \$4.50, while the variable cost of



producing a pepperoni-and-cheese pizza was \$5.00. All pizzas were assumed to be sold for \$9.00, regardless of type or demand level.

2. SUMMARY OF RESULTS

After careful analysis, a range of expected monetary values were calculated for hypothetical production and demand levels using the known probability distributions for cheese and pepperoni-and-cheese pizzas. Arrowmark Vending should avoid overproducing pizzas, as excess production increases the risk of losses when demand is low. Producing 400 pizzas minimizes downside risk but limits potential profit when demand is high, while producing 600 pizzas increases exposure to losses while demand is low. Based on the expected value calculations, producing 500 pizzas of each type yields the highest expected profit over the season. The total expected profit for plain cheese and pepperoni pizzas is \$1075 and \$1140, respectively. While low-demand outcomes do pose risk, the higher expected profitability supports the recommendation to produce 500 pizzas of each type per game. Therefore, Tom Kealey can confidently recommend producing 500 pizzas of each type, as Arrowmark Vending can expect positive profitability over the course of the season, even if demand is low on the occasional home game.

Table 1 Production vs Projected EMV

Production	Cheese Expected Profit (EMV)	Pepp Expected Profit (EMV)
200	\$400.00	N/A
300	\$760.00	\$700.00
400	\$985.00	\$1,010.00
500	\$1,075.00	\$1,140.00
600	\$985.00	\$1,045.00
700	\$715.00	\$725.00
800	\$355.00	\$270.00
900	\$(50.00)	N/A



3. ANALYSIS

The analysis evaluates the expected monetary value (EMV) associated with each pizza production level by examining projected costs and revenues across all possible demand scenarios. For each production quantity, the gross profit corresponding to each demand level is calculated and then weighted by the probability that the demand level occurs. These probability-weighted profits represent the expected value of each individual demand scenario.

The total EMV for a given production decision is obtained by summing the expected values across all demand scenarios and subtracting the fixed production cost, which is incurred regardless of demand. This approach allows for a direct comparison of production alternatives based on expected profitability while explicitly accounting for demand uncertainty.

$$\text{Expected Profit}(Q) = \sum [P(D_i) \times (\text{Revenue}(Q, D_i) - \text{Variable Cost}(Q))] - \text{Fixed Cost}$$

Figure 1 Expected Profit Formula

D_i = demand level i

$P(D_i)$ = probability that demand level i occurs

Revenue (Q, D_i) = revenue earned when producing Q units and demand is D_i

Variable Cost(Q) = total variable cost of producing Q units

Fixed Cost = fixed production cost (independent of demand)

Figure 2 Expected Profit Variables

Snippet of EMV Model: If 300 Pepperoni Pizzas Are Produced								
Pizzas Produced	Pepp Demand	Pizza Sold	Gross Revenue	Fixed Cost	Variable Cost	Gross Profit	Probability	Expected Profit Adjusted for Probability
300	300	300	\$2700.00	\$500.00	\$1,500.00	\$700.00	0.1	\$70.00
300	400	300	\$2,700.00	\$500.00	\$1,500.00	\$700.00	0.2	\$140.00
300	500	300	\$2,700.00	\$500.00	\$1,500.00	\$700.00	0.25	\$175.00
300	600	300	\$2,700.00	\$500.00	\$1,500.00	\$700.00	0.25	\$175.00
300	700	300	\$2,700.00	\$500.00	\$1,500.00	\$700.00	0.15	\$105.00
300	800	300	\$2,700.00	\$500.00	\$1,500.00	\$700.00	0.05	\$35.00

Figure 3 EMV Model for 300 Pepp Pizzas

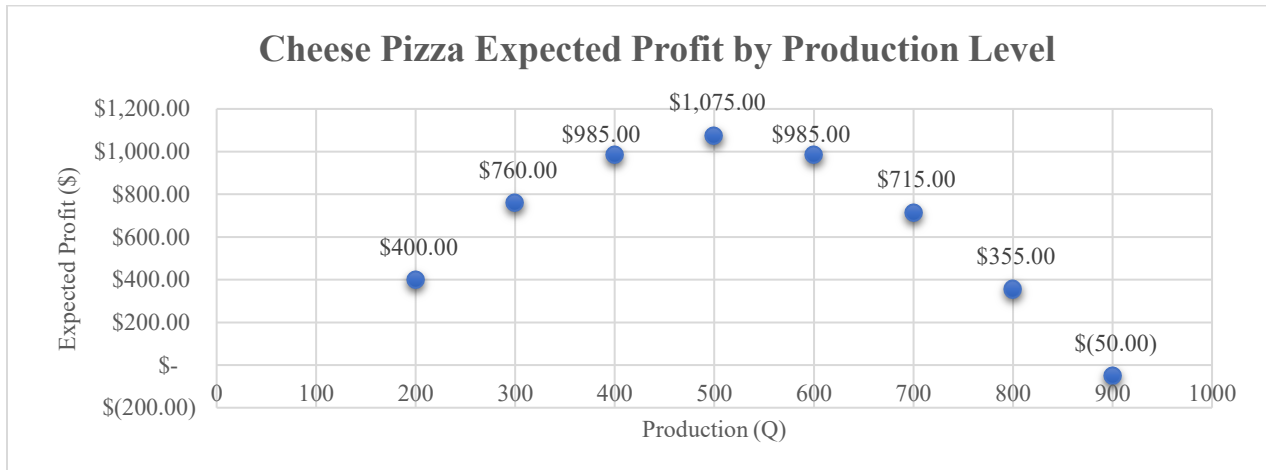


Figure 4 Cheese pizza expected profit by production

As seen in table 1 and figure 4 above, expected profit for cheese pizza increases steadily as production rises from 200 to about 500 units, then declines after that point. Profit starts around \$400 at 200 pizzas, climbs to about \$760 at 300, \$985 at 400, and peaks near \$1,075 at 500 pizzas. Beyond 500, profit drops to \$985 at 600, \$715 at 700, and \$355 at 800, turning into a small loss of about \$50 at 900.

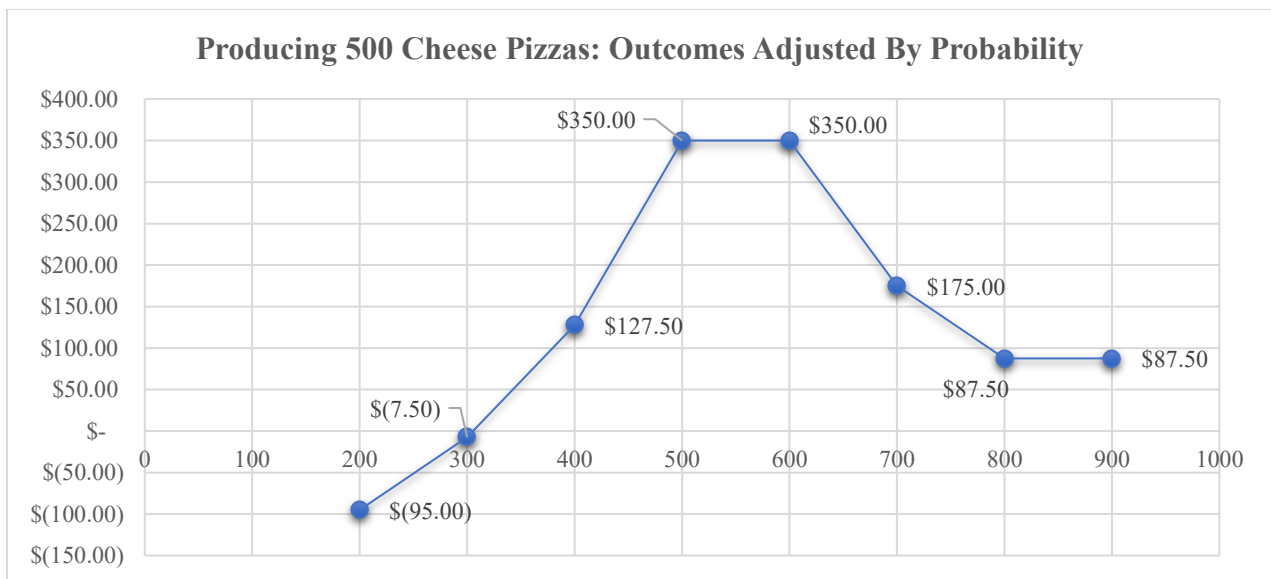


Figure 5 Expected profit adjusted by probability if 500 Cheese Pizzas are produced

Figure 5 shows the probability of adjusted profit for a fixed production level of 500 pizzas across demand outcomes. Profit is negative at low demand because unsold units create waste and reduce returns. Expected profit increases through the mid-demand range and peaks near \$350 at



500 to 600 orders, where production and realized sales are closely aligned. Profit then levels off and declines beyond this range due to the 500 unit production cap and lower probability of higher order volumes.

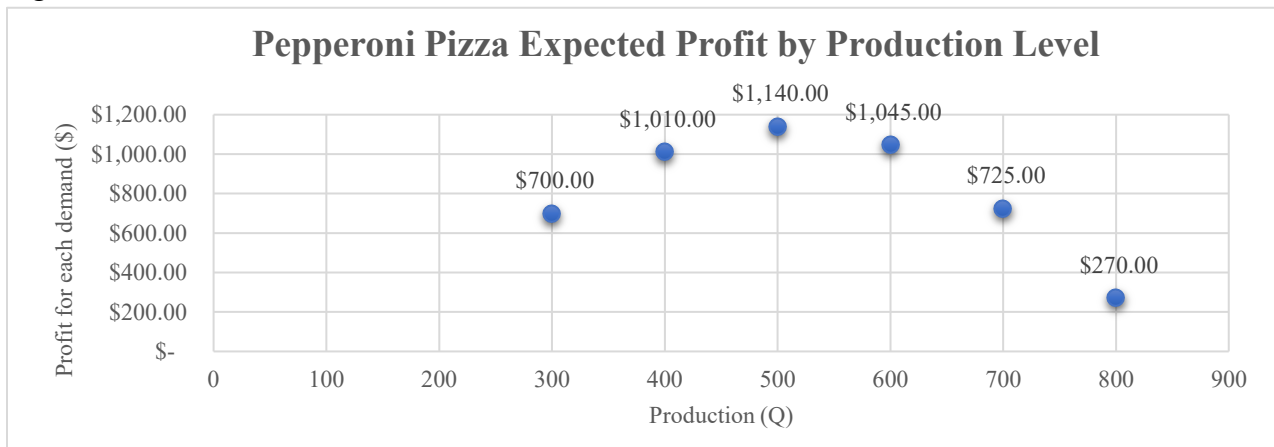


Figure 6 Pepperoni pizza expected profit versus production

As seen in table 1 and figure 6 above, expected profit rises as pepperoni pizza production increases from 300 to about 500 units, then declines beyond that point. Profit is approx. \$700 at 300 pizzas, grows to \$1,010 at 400, and peaks near \$1,140 at 500 pizzas. After that, profit drops slightly to about \$1,045 at 600, then falls more sharply to around \$725 at 700 and \$270 at 800.

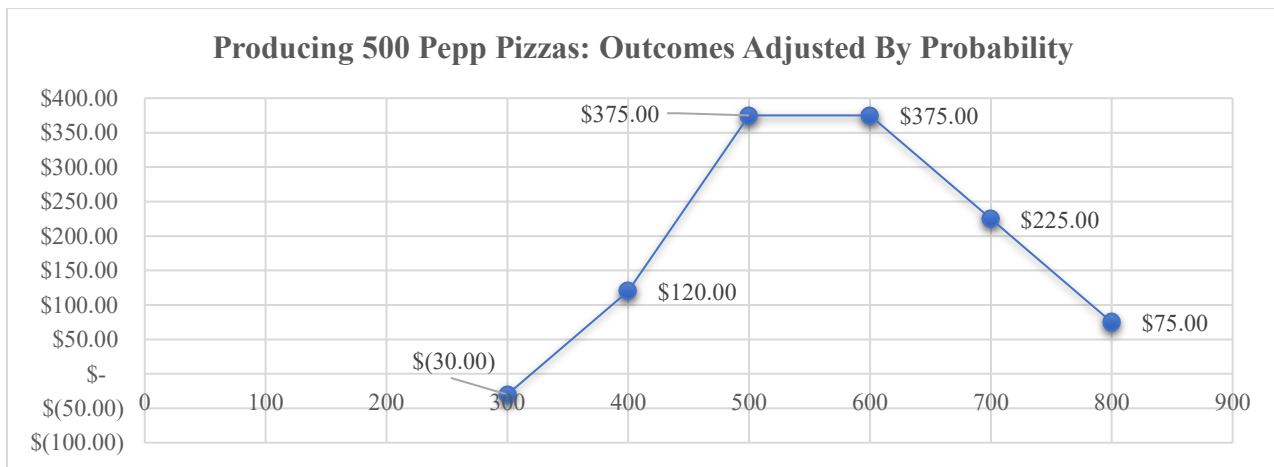


Figure 7 Expected profit adjusted by probability if 500 Pepperoni Pizzas are produced

Figure 7 shows the probability adjusted profit curve is negative at low demand, about (\$30) at 300 orders, due to excess inventory. Expected profit increases to a maximum near \$375 at 500 to 600 orders, where production and realized sales align. Profit then declines to about \$225 at 700 and \$75 at 800 as demand moves away from the production level.



4. RECOMMENDATION

Based on the analysis and summary of results, it is recommended that Tom Kealey, operations manager for Arrowmark, plan to have 500 cheese and 500 pepperoni pizzas available for Purdue football games. This allocation provides the highest expected profit given demand uncertainty: $\$1,075$ (cheese) + $\$1,140$ (pepperoni) = $\$2,215$ total expected profit.

The probability that demand exceeds 500 for cheese pizzas (stockout risk) is 40%. The probability that demand exceeds 500 for pepperoni pizzas is 45%. Pepperoni is slightly more likely to run out, but accepting these stockout risks maximizes expected profit compared to overstocking.

The probability of overage (demand below 500) is 40% for cheese and 30% for pepperoni. While leftovers will be donated to the local homeless shelter, the cost of unsold inventory makes overstocking less profitable than the recommended quantities. The 500-unit order level optimally balances these overage and underage risks for both pizza types.

5. ACTION PLAN

Tom Kealey, operations manager for Arrowmark, should order 500 cheese and 500 pepperoni pizzas for each home football game.

If ticket sales or weather forecasts suggest significantly different attendance, demand probabilities should be updated and the model re-run before placing orders.

As additional home games occur this season, actual demand data should be collected and analyzed to refine the probability distributions. If observed demand patterns differ significantly from current estimates, optimal order quantities should be recalculated to improve profitability for remaining games.

This action plan assumes that cheese and pepperoni demands are independent. It should be noted that in practice, rather than simply not buying pizza if their preferred type were out of stock, a customer would buy the alternative. This expected consumer behavior would reduce actual stockout losses, which in turn makes our analysis conservative.